

ModE Project 5 - Group 7

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1 Problem Formulations

1.1 MILP

1.1.1 Objective

$$\min_{\mathbf{x}} \quad c_G \sum_k E_{\text{Gas},k} + c_{\text{el}} \sum_k E_{\text{el},k}$$

with

$$\begin{aligned} E_{\text{el},k} &= \Delta t \cdot P_{\text{grid},k} \\ E_{\text{Gas},k} &= \Delta t \left[\sum_{i \in \mathcal{B}} \dot{Q}_{\text{B},i,k}^{\text{in}} + \sum_{i \in \mathcal{CHP}} \dot{Q}_{\text{CHP},i,k}^{\text{in}} \right] \end{aligned}$$

1.1.2 Decision variables

At each time step $k \in \{1, \dots, N\}$, the decision variable vector of the independent variables is

$$\tilde{\mathbf{x}}_k = \left[\dot{Q}_{\text{B},1,k}^{\text{in}}, \dot{Q}_{\text{B},2,k}^{\text{in}}, \dot{Q}_{\text{CHP},1,k}^{\text{in}}, \dot{Q}_{\text{CHP},2,k}^{\text{in}}, \dot{Q}_{\text{TES},k}^{\text{in}}, \dot{Q}_{\text{TES},k}^{\text{out}}, \right. \\ \left. P_{\text{grid},k}, \delta_{\text{B},1,k}, \delta_{\text{B},2,k}, \delta_{\text{CHP},1,k}, \delta_{\text{CHP},2,k}, \delta_{\text{TES},k}^{\text{in}}, \delta_{\text{TES},k}^{\text{out}} \right]^{\top}$$

With all variables being continuous and contained in $\mathbb{R}$ except the binary variables $\delta \in \{0, 1\}$.

The auxiliary variables that could be eliminated by substitution (to get a dense formulation) are at each time step $k \in \{1, \dots, N\}$:

$$\mathbf{y}_k = \left[E_{\text{TES},k}, \dot{Q}_{\text{B},1,k}^{\text{out}}, \dot{Q}_{\text{B},2,k}^{\text{out}}, \dot{Q}_{\text{CHP},1,k}^{\text{out}}, \dot{Q}_{\text{CHP},2,k}^{\text{out}}, \right. \\ \left. P_{\text{CHP},1,k}^{\text{out}}, P_{\text{CHP},2,k}^{\text{out}}, E_{\text{G},k}, E_{\text{el},k} \right]^{\top}$$

They are kept as decision variables for the sparse formulation. The full decision variable vector thus is:

$$\mathbf{x}_k = \begin{bmatrix} \tilde{\mathbf{x}}_k^{\top} & \mathbf{y}_k^{\top} \end{bmatrix}^{\top}.$$

Time discretization The continuous time horizon is discretized into N uniform time steps indexed by $k \in \{0, 1, \dots, N\}$, with

$$t \in [0, t_f], \quad t_k = k \cdot \Delta t, \quad \Delta t = t_{k+1} - t_k \quad \forall k, \quad t_f = N \Delta t$$

The demand and outputs of the conversion components are assumed constant throughout each time period (zero-order hold discretization). The differential equation **TES(I)** is discretized exactly using the matrix exponential. This yields the discrete heat content of the TES $E_{\text{TES},k}$.

1.1.3 Constraints

Subject to:

TES

$$(I) \quad \dot{E}_{\text{TES}}(t) = \eta_{\text{TES}}^{\text{in}} \dot{Q}_{\text{TES}}^{\text{in}}(t) - \frac{\dot{Q}_{\text{TES}}^{\text{out}}(t)}{\eta_{\text{TES}}^{\text{out}}} - \frac{E_{\text{TES}}(t)}{\tau_{\text{loss}}}$$

discretized, see 2.1: $E_{\text{TES},k+1} = a E_{\text{TES},k} + b_1 \dot{Q}_{\text{TES},k}^{\text{in}} + b_2 \dot{Q}_{\text{TES},k}^{\text{out}}$

$$(II) \quad E_{\text{TES},k} \leq E_{\text{TES}}^{\text{nom}}$$

$$(III) \quad -E_{\text{TES},k} \leq -E_{\text{TES}}^{\text{min}}$$

$$(IV) \quad \dot{Q}_{\text{TES},k}^{\text{in}} \leq \delta_{\text{TES},k}^{\text{in}} \frac{E_{\text{TES}}^{\text{nom}}}{\tau_{\text{in}}}$$

$$(V) \quad -\dot{Q}_{\text{TES},k}^{\text{in}} \leq -\delta_{\text{TES},k}^{\text{in}} \dot{Q}_{\text{TES}}^{\text{in,min}}$$

$$(VI) \quad \dot{Q}_{\text{TES},k}^{\text{out}} \leq \delta_{\text{TES},k}^{\text{out}} \frac{E_{\text{TES}}^{\text{nom}}}{\tau_{\text{out}}}$$

$$(VII) \quad -\dot{Q}_{\text{TES},k}^{\text{out}} \leq -\delta_{\text{TES},k}^{\text{out}} \dot{Q}_{\text{TES}}^{\text{out,min}}$$

$$(VIII) \quad E_{\text{TES}}(t=0) = E_{\text{TES}}(t=t_f) \quad \text{Cycle constraint}$$

$$(IX) \quad E_{\text{TES}}(t=0) = E_{\text{TES},0} \quad \text{Initial condition}$$

$$(X) \quad \delta_{\text{TES},k}^{\text{in}} + \delta_{\text{TES},k}^{\text{out}} \leq 1$$

$\forall t \in [0, t_f], \forall k = 1, \dots, N$. Note that the most likely case is that the TES can be discharged fully and that there are no minimum charging or discharging fluxes, then $E_{\text{TES}}^{\text{min}} = \dot{Q}_{\text{TES}}^{\text{in,min}} = \dot{Q}_{\text{TES}}^{\text{out,min}} = 0$.

Boiler B_i

$$(I) \quad \dot{Q}_{B,i,k}^{\text{out}} = \dot{Q}_B^{\text{out,nom}} \left[\delta_{B,i,k} \lambda_B^{\text{out,min}} + \frac{1}{\beta_B} \left(\frac{\dot{Q}_{B,i,k}^{\text{in}} \eta_B^{\text{nom}}}{\dot{Q}_B^{\text{out,nom}}} - \delta_{B,i,k} \lambda_B^{\text{in,min}} \right) \right]$$

$$(II) \quad \dot{Q}_{B,i,k}^{\text{in}} \leq \delta_{B,i,k} \frac{\dot{Q}_B^{\text{out,nom}}}{\eta_B^{\text{nom}}}$$

$$(III) \quad -\dot{Q}_{B,i,k}^{\text{in}} \leq -\delta_{B,i,k} \lambda_B^{\text{in,min}} \frac{\dot{Q}_B^{\text{out,nom}}}{\eta_B^{\text{nom}}}$$

$$\forall k = 1, \dots, N, \quad \forall i \in \mathcal{B} = \{1, 2\}$$

CHP_i

$$\begin{aligned}
\text{(I)} \quad & \dot{Q}_{\text{CHP},i,k}^{\text{out}} = \dot{Q}_{\text{CHP}}^{\text{out,nom}} \left[\delta_{\text{CHP},i,k} \lambda_{\text{CHP,th}}^{\text{out,min}} + \frac{1}{\beta_{\text{CHP,th}}} \left(\frac{\dot{Q}_{\text{CHP},i,k}^{\text{in}} \eta_{\text{CHP,th}}^{\text{nom}}}{\dot{Q}_{\text{CHP}}^{\text{out,nom}}} - \delta_{\text{CHP},i,k} \lambda_{\text{CHP}}^{\text{in,min}} \right) \right] \\
\text{(II)} \quad & P_{\text{CHP},i,k}^{\text{out}} = P_{\text{CHP}}^{\text{out,nom}} \left[\delta_{\text{CHP},i,k} \lambda_{\text{CHP,el}}^{\text{out,min}} + \frac{1}{\beta_{\text{CHP,el}}} \left(\frac{\dot{Q}_{\text{CHP},i,k}^{\text{in}} \eta_{\text{CHP,el}}^{\text{nom}}}{P_{\text{CHP}}^{\text{out,nom}}} - \delta_{\text{CHP},i,k} \lambda_{\text{CHP}}^{\text{in,min}} \right) \right] \\
\text{(III)} \quad & \dot{Q}_{\text{CHP},i,k}^{\text{in}} \leq \delta_{\text{CHP},i,k} \frac{\dot{Q}_{\text{CHP}}^{\text{out,nom}}}{\eta_{\text{CHP,th}}^{\text{nom}}} \\
\text{(IV)} \quad & -\dot{Q}_{\text{CHP},i,k}^{\text{in}} \leq -\delta_{\text{CHP},i,k} \lambda_{\text{CHP}}^{\text{in,min}} \frac{P_{\text{CHP}}^{\text{out,nom}}}{\eta_{\text{CHP,el}}^{\text{nom}}}
\end{aligned}$$

$$\forall k = 1, \dots, N, \quad \forall i \in \mathcal{CHP} = \{1, 2\}$$

Demand Note that the TES being continuous time but the demand only being given in discrete steps, the actual heat delivered by the TES is first integrated and then averaged.

$$\begin{aligned}
\text{(I)} \quad & \dot{Q}_{\text{D},k} = \sum_{i \in \mathcal{CHP}} \dot{Q}_{\text{CHP},i,k}^{\text{out}} + \sum_{i \in \mathcal{B}} \dot{Q}_{\text{B},i,k}^{\text{out}} + \dot{Q}_{\text{TES},k}^{\text{out}} - \dot{Q}_{\text{TES},k}^{\text{in}} \\
\text{(II)} \quad & P_{\text{D},k} = \sum_{i \in \mathcal{CHP}} P_{\text{CHP},i,k}^{\text{out}} + P_{\text{grid},k}
\end{aligned}$$

$$\forall k = 1, \dots, N$$

1.2 Approximated LP

The first LP formulation is obtained by removing the binary variables from the decision variables and the constraints. The objective and time discretization stay the same.

Decision variables At each time step $k \in \{1, \dots, N\}$, the decision variable vector of the independent variables is

$$\tilde{\mathbf{x}}_k = \left[\dot{Q}_{\text{B},1,k}^{\text{in}}, \dot{Q}_{\text{B},2,k}^{\text{in}}, \dot{Q}_{\text{CHP},1,k}^{\text{in}}, \dot{Q}_{\text{CHP},2,k}^{\text{in}}, \dot{Q}_{\text{TES},k}^{\text{in}}, \dot{Q}_{\text{TES},k}^{\text{out}}, P_{\text{grid},k} \right]$$

With all variables being continuous and in $\mathbb{R}^n$. The auxiliary variables stay the same.

1.2.1 Constraints

The constraints of the production components and of the TES change, as described in more detail in 2.2. The demand constraints stay the same.

Subject to:

TES

- (I) $E_{\text{TES},k} = a E_{\text{TES},k-1} + b_1 \dot{Q}_{\text{TES},k-1}^{\text{in}} + b_2 \dot{Q}_{\text{TES},k-1}^{\text{out}}$
- (II) $E_{\text{TES},k} \leq E_{\text{TES}}^{\text{nom}}$
- (III) $-E_{\text{TES},k} \leq -E_{\text{TES}}^{\text{min}}$
- (IV) $\dot{Q}_{\text{TES},k}^{\text{in}} \leq \frac{E_{\text{TES}}^{\text{nom}}}{\tau^{\text{in}}}$
- (V) $-\dot{Q}_{\text{TES},k}^{\text{in}} \leq -\dot{Q}_{\text{TES}}^{\text{in,min}}$
- (VI) $\dot{Q}_{\text{TES},k}^{\text{out}} \leq \frac{E_{\text{TES}}^{\text{nom}}}{\tau^{\text{out}}}$
- (VII) $-\dot{Q}_{\text{TES},k}^{\text{out}} \leq -\dot{Q}_{\text{TES}}^{\text{out,min}}$
- (VIII) $E_{\text{TES}}(t=0) = E_{\text{TES}}(t=t_f)$ Cycle constraint
- (IX) $E_{\text{TES}}(t=0) = E_{\text{TES},0}$ Initial condition

$$\forall t \in [0, t_f], \forall k = 1, \dots, N.$$

Boiler B_i

- (I) $\dot{Q}_{B,i,k}^{\text{out}} = \alpha_B \dot{Q}_{B,i,k}^{\text{in}}$
- (II) $\dot{Q}_{B,i,k}^{\text{in}} \leq \frac{\dot{Q}_B^{\text{out,nom}}}{\eta_B^{\text{nom}}}$
- (III) $-\dot{Q}_{B,i,k}^{\text{in}} \leq 0$

$$\forall k = 1, \dots, N, \quad \forall i \in \mathcal{B} = \{1, 2\}$$

CHP $_i$

- (I) $\dot{Q}_{\text{CHP},i,k}^{\text{out}} = \alpha_{\text{CHP,th}} \dot{Q}_{\text{CHP},i,k}^{\text{in}}$
- (II) $P_{\text{CHP},i,k}^{\text{out}} = \alpha_{\text{CHP,el}} \dot{Q}_{\text{CHP},i,k}^{\text{in}}$
- (III) $\dot{Q}_{\text{CHP},i,k}^{\text{in}} \leq \frac{\dot{Q}_{\text{CHP}}^{\text{out,nom}}}{\eta_{\text{CHP,th}}^{\text{nom}}}$
- (IV) $-\dot{Q}_{\text{CHP},i,k}^{\text{in}} \leq 0$

$$\forall k = 1, \dots, N, \quad \forall i \in \mathcal{CHP} = \{1, 2\}$$

1.3 Relaxed LP

The simpler definition of the LP is the continuous relaxation of the binary MILP, as is done in the first iteration of Branch&Bound. This LP is guaranteed to yield a lower bound on the OPEX of the MILP.

To do this, the binary variables are changed from $\delta \in \{0, 1\}$ to $\delta \in [0, 1]$.

1.4 Parameters

β is the slope of the part-load efficiency curve for the boilers and the electrical and thermal outputs of the CHP respectively.

$$\beta_j = \frac{1 - \lambda_j^{\text{in},\text{min}}}{1 - \lambda_j^{\text{out},\text{min}}}, \quad j \in \{\text{B}, \text{CHP}_{\text{th}}, \text{CHP}_{\text{el}}\}$$

Other parameters have the pre-defined numeric values shown below.

General $\Delta t = 1 \text{ h}$

$t_f = 168 \text{ h}$ (7 days)

c_G, c_{el} uncertain parameters sampled

within ranges $[R_G^+, R_G^-]$ and $[R_{\text{el}}^+, R_{\text{el}}^-]$

TES $\tau^{\text{loss}} = 200 \text{ h}$

$\tau^{\text{in}} = \tau^{\text{out}} = 1 \text{ h}$

$\eta_{\text{TES}}^{\text{in}} = \eta_{\text{TES}}^{\text{out}} = 0.95$

$a \approx 0.995012$

$b_1 \approx 0.94763 \text{ h}$

$b_2 \approx -1.05 \text{ h}$

$E_{\text{TES}}^{\text{nom}} = 1000 \text{ kWh}$

$E_{\text{TES}}^{\text{min}} = \dot{Q}_{\text{TES}}^{\text{in},\text{min}} = \dot{Q}_{\text{TES}}^{\text{out},\text{min}} = 0$

$E_{\text{TES},0} = 0 \text{ kWh}$

Boiler $\dot{Q}_{\text{B}}^{\text{out},\text{nom}} = 530 \text{ kW}$

$\eta_{\text{B}}^{\text{nom}} = 0.8$

$\lambda_{\text{B}}^{\text{in},\text{min}} = 0.173$

$\lambda_{\text{B}}^{\text{out},\text{min}} = 0.2$

CHP $\dot{Q}_{\text{CHP}}^{\text{out},\text{nom}} = 470 \text{ kW}$

$P_{\text{CHP}}^{\text{out},\text{nom}} = 380 \text{ kW}$

$\eta_{\text{CHP,th}}^{\text{nom}} = 0.481$

$\eta_{\text{CHP,el}}^{\text{nom}} = 0.389$

$\lambda_{\text{CHP,th}}^{\text{in},\text{min}} = \lambda_{\text{CHP,el}}^{\text{in},\text{min}} = 0.582$

$\lambda_{\text{CHP,th}}^{\text{out},\text{min}} = 0.622$

$\lambda_{\text{CHP,el}}^{\text{out},\text{min}} = 0.5$

Demand $\dot{Q}_{\text{D},k}, P_{\text{D},k}$ taken from the given time series of the demand (deterministic)

2 Details

2.1 Discretization of TES Equations

The heat balance equation of the TES **TES(I)** is a **linear** first-order ordinary differential equation. Because of its linearity, it can be rewritten using the state space formulation as follows:

$$\begin{aligned}\dot{\mathbf{x}}(t) &= A \mathbf{x}(t) + B \mathbf{u}(t) \\ \dot{\mathbf{x}}(t) &= \dot{E}_{\text{TES}}(t) \\ \mathbf{x}(t) &= E_{\text{TES}}(t) \\ \mathbf{u}(t) &= \begin{bmatrix} \dot{Q}_{\text{TES}}^{\text{in}}(t) & \dot{Q}_{\text{TES}}^{\text{out}}(t) \end{bmatrix}^{\top} \\ A &= -\frac{1}{\tau_{\text{loss}}} \\ B &= \begin{bmatrix} \eta_{\text{TES}}^{\text{in}} & -\frac{1}{\eta_{\text{TES}}^{\text{out}}} \end{bmatrix}\end{aligned}$$

Linear state space models can be **exactly** discretized using the matrix exponential equations, assuming that the input is discretized with zero-order hold. This is a valid assumption, as we want to define the *decision variables* $\mathbf{u}(t)$ anyways ideally only for every hour.

The theoretical discretization of the linear state space equations is as follows:

$$\begin{aligned}\mathbf{x}_{k+1} &= A_d \mathbf{x}_k + B_d \mathbf{u}_k \\ A_d &= e^{A\Delta t} \\ B_d &= B \cdot \int_0^{\Delta t} e^{A\tau} d\tau\end{aligned}$$

Evaluating these equations, the following values are obtained:

$$\begin{aligned}A_d &\approx 0.99501 \\ B_d &\approx [0.94763, -1.05]\end{aligned}$$

Comparing this with explicit Euler discretization which would use $A_d^E = I_{1 \times 1} + A\Delta t = 0.995$ and $B_d^E = B \Delta t \approx [0.95, -1.0526]$, the relative errors (per component) are: $\frac{A_d - A_d^E}{A_d} \approx 1.2542 \cdot 10^{-5}$ and $\frac{B_d - B_d^E}{B_d} \approx [-0.0025021, -0.0025021]$. The errors are small due to the large time scale differences between the discharge of the TES and the discretization time-step.

If one were to increase the discretization time-step to e.g. $\Delta t = 10$ h, the errors would increase to $\frac{A_d - A_d^E}{A_d} \approx 0.0012925$ and $\frac{B_d - B_d^E}{B_d} \approx [-0.025208, -0.025208]$. The increase in the relative error is of multiple orders of magnitude for A_d^E and one order of magnitude for B_d^E .

The simulated homogeneous response of the TES upon being fully charged is shown in Fig. 1 for both methods and both time-steps.

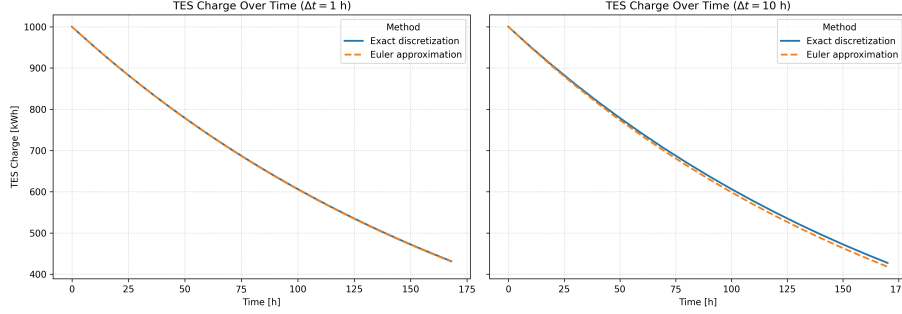


Figure 1: Comparison of discretization methods for the TES differential equation. Plots show the homogeneous response of the TES upon fully charged with $E_{\text{TES}}(t_0) = 1000 \text{ kWh}$ at the beginning.

2.2 Linearization of constraints

The part-load behavior of the production components (Boiler and CHP) contains binary variables. These can be eliminated to get a linear formulation, as is done for the Approximated LP. This implicitly removes the minimum operating point.

The linearization of the production components is obtained by fitting a linear function through the original piece-wise linear efficiency curve. This is done numerically through optimization in `scipy`. This is the minimization problem that is solved:

$$\begin{aligned}
 & \min_{\alpha_i} \int (\alpha_i x_i - y_i)^2 dx_i \\
 \text{s.t. } & y_i = \begin{cases} y_i^{\text{nom}} \left[\lambda_i^{\text{y,min}} + \frac{1}{\beta_i} \left(\frac{x_i \eta_i^{\text{nom}}}{y_i^{\text{nom}}} - \lambda_i^{\text{x,min}} \right) \right] & \text{if } x_i \geq x_i^{\text{min}} \\ 0 & \text{otherwise} \end{cases} \\
 & x_i \in [0, x_i^{\text{max}}] \\
 & i = \text{B, CHP}_{\text{el}}, \text{CHP}_{\text{th}}
 \end{aligned}$$

The objective is equivalent to minimizing the mean squared error between the two functions. This optimization results in the following constants:

$$\alpha_{\text{B}} = 0.8078 \quad (1)$$

$$\alpha_{\text{CHP,th}} = 0.3948 \quad (2)$$

$$\alpha_{\text{CHP,el}} = 0.298 \quad (3)$$

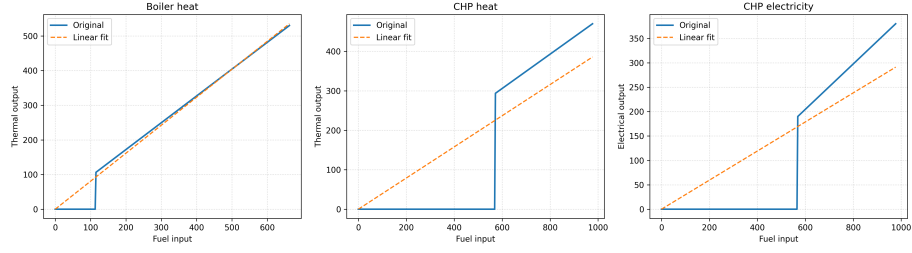


Figure 2: Linearization results for the 3 components.

See Fig. 2 for the visualization of the new operating behavior of the production components.

The linearization of the TES is achieved by removing the restriction of no simultaneous charge and discharge. In theory, the solver can thus choose to cycle energy through the TES to dissipate heat. However, as it can now control the heat output of the CHP and Boiler freely, this would only incur additional costs.